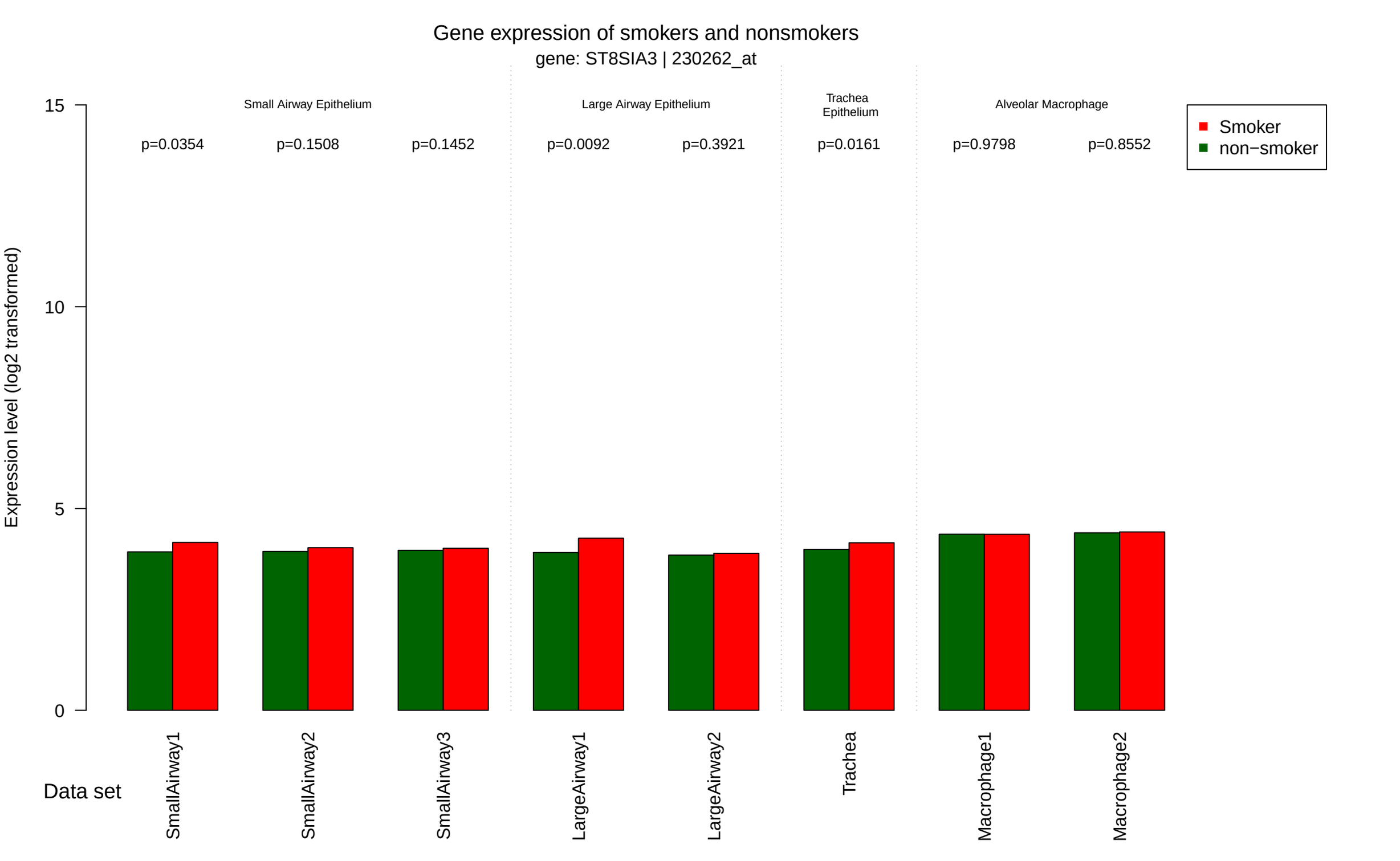
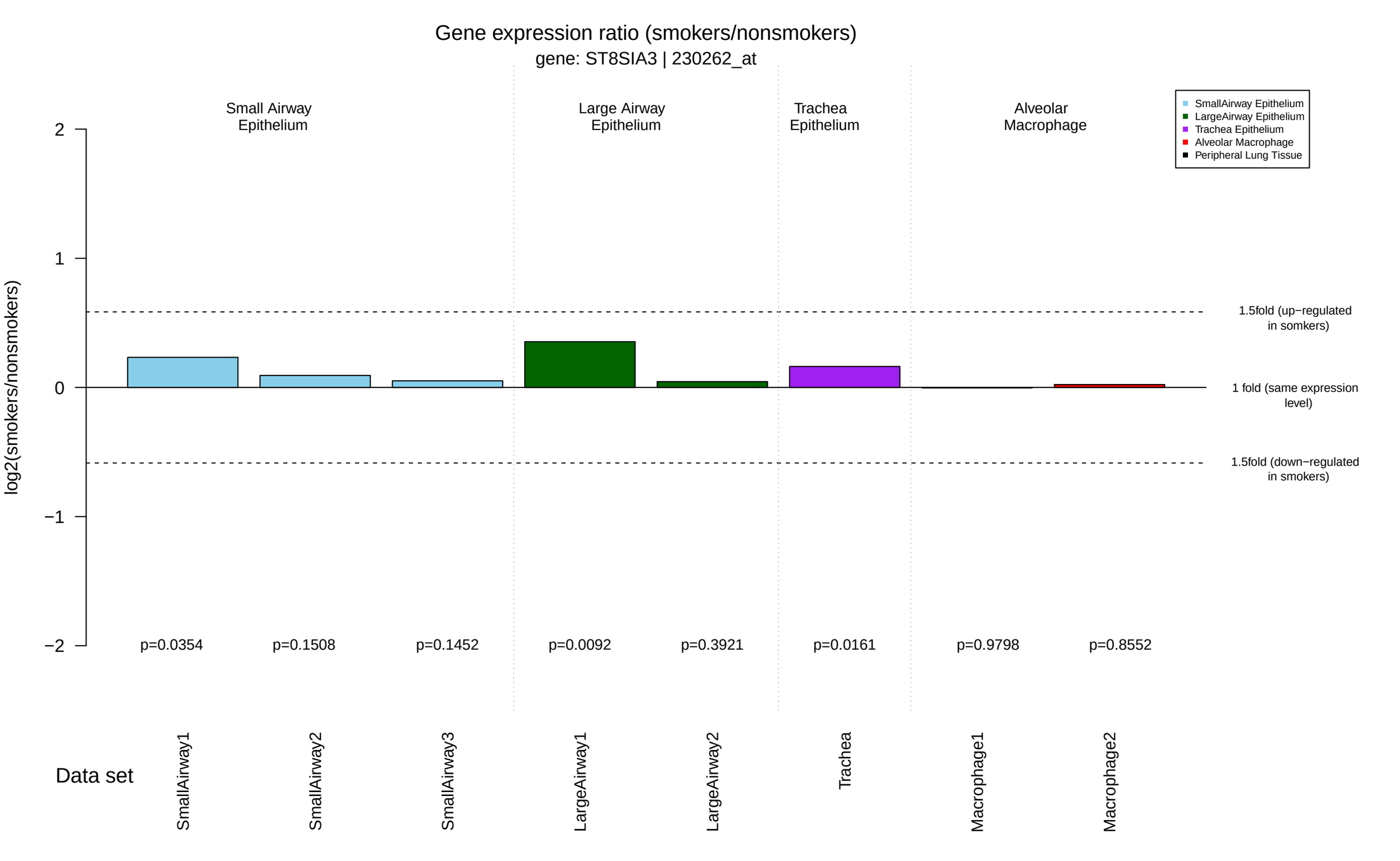
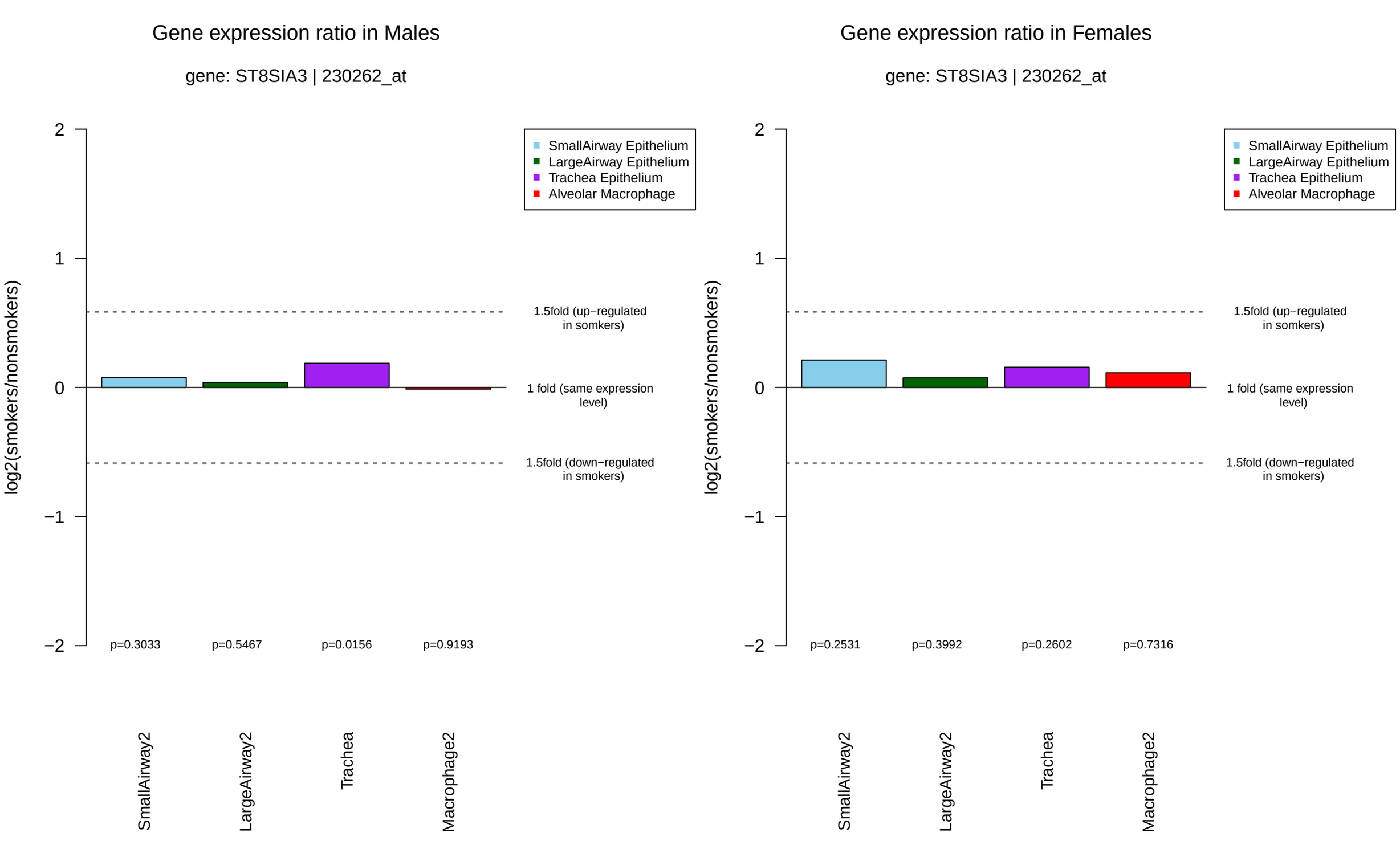
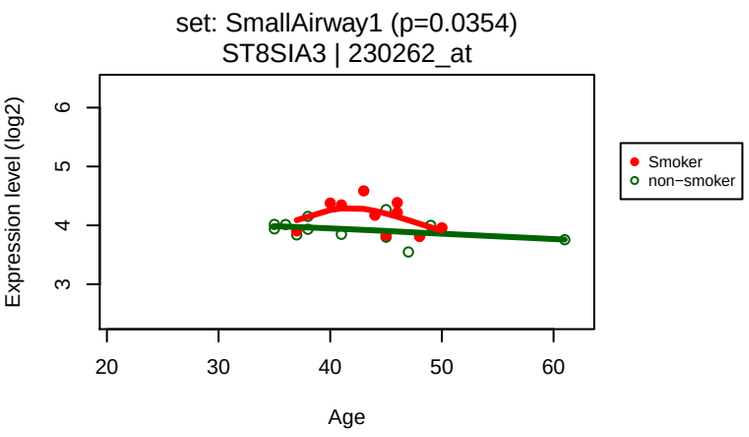
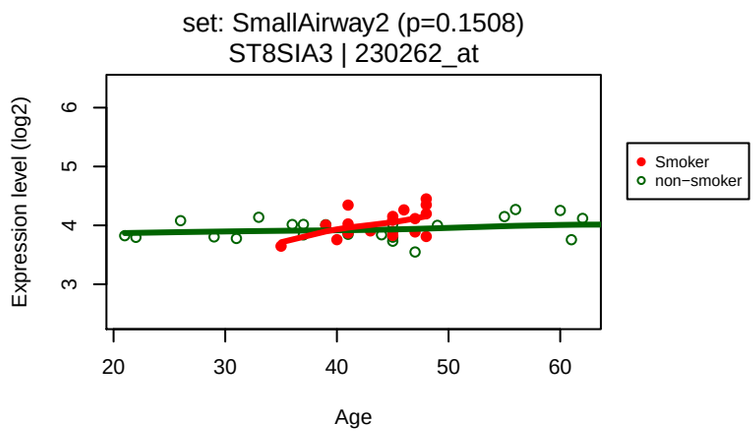
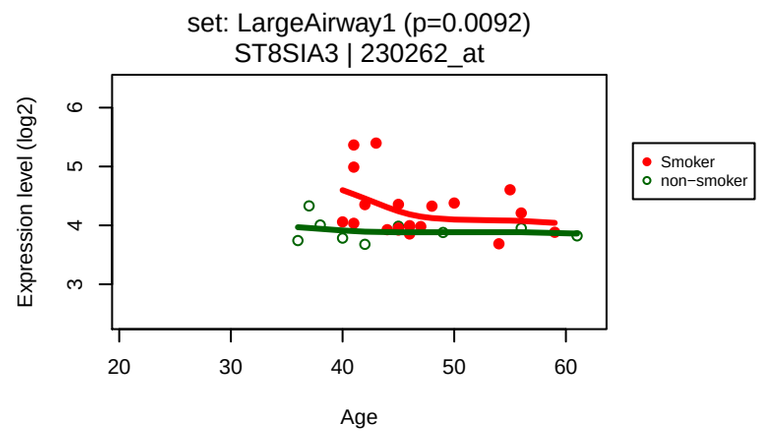

Gene expression of smokers and nonsmokers
gene: ST8SIA3 | 230262_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0354
p=0.1508
p=0.1452
p=0.0092
p=0.3921
p=0.0161
p=0.9798
p=0.8552
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

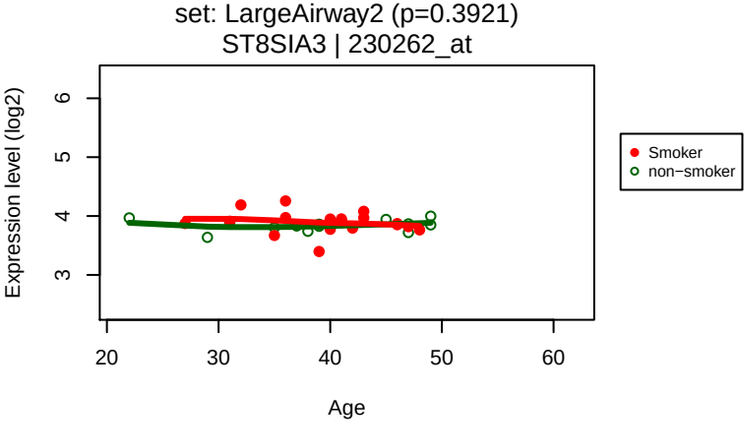

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA3 | 230262_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0354
p=0.1508
p=0.1452
p=0.0092
p=0.3921
p=0.0161
p=0.9798
p=0.8552
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

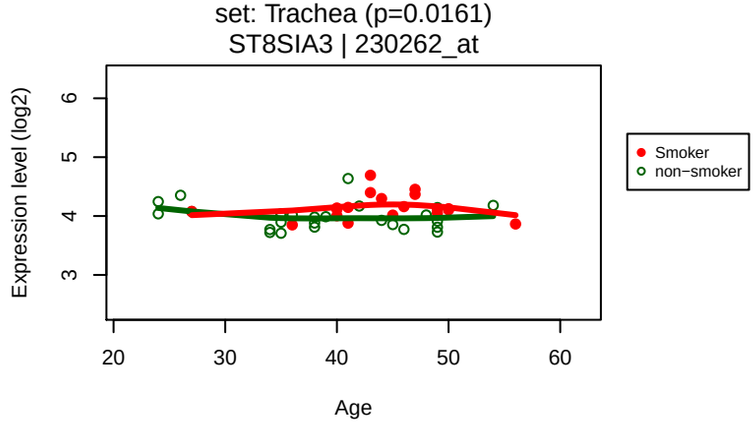

Gene expression ratio in Males
gene: ST8SIA3 | 230262_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3033
p=0.5467
p=0.0156
p=0.9193
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA3 | 230262_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2531
p=0.3992
p=0.2602
p=0.7316
SmallAirway2
LargeAirway2
Trachea
Macrophage2

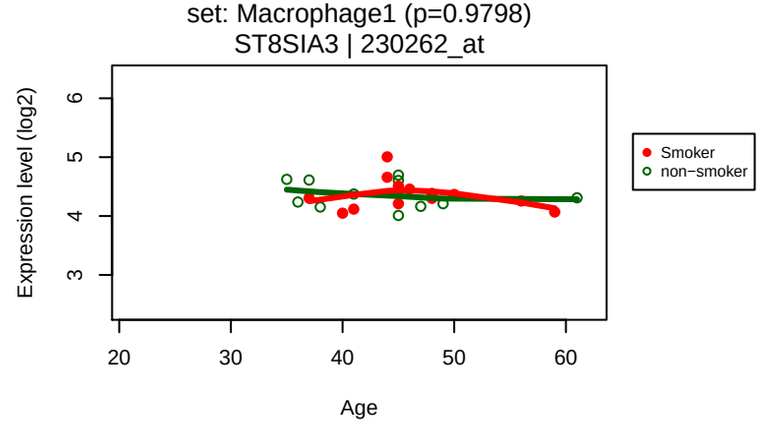

set: SmallAirway1 (p=0.0354)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker

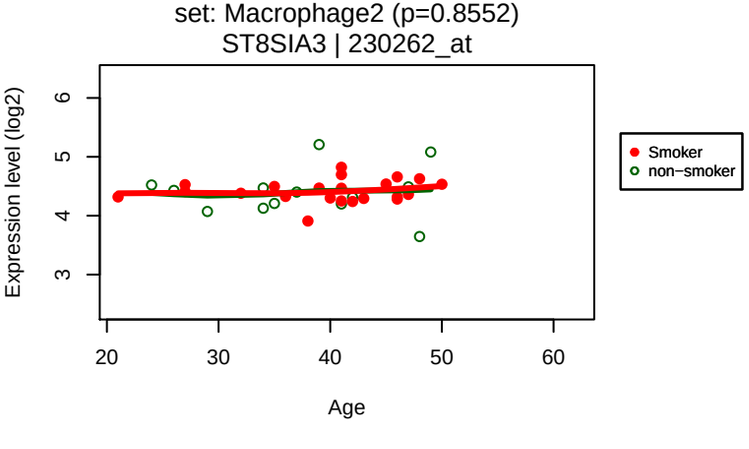

set: SmallAirway2 (p=0.1508)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker

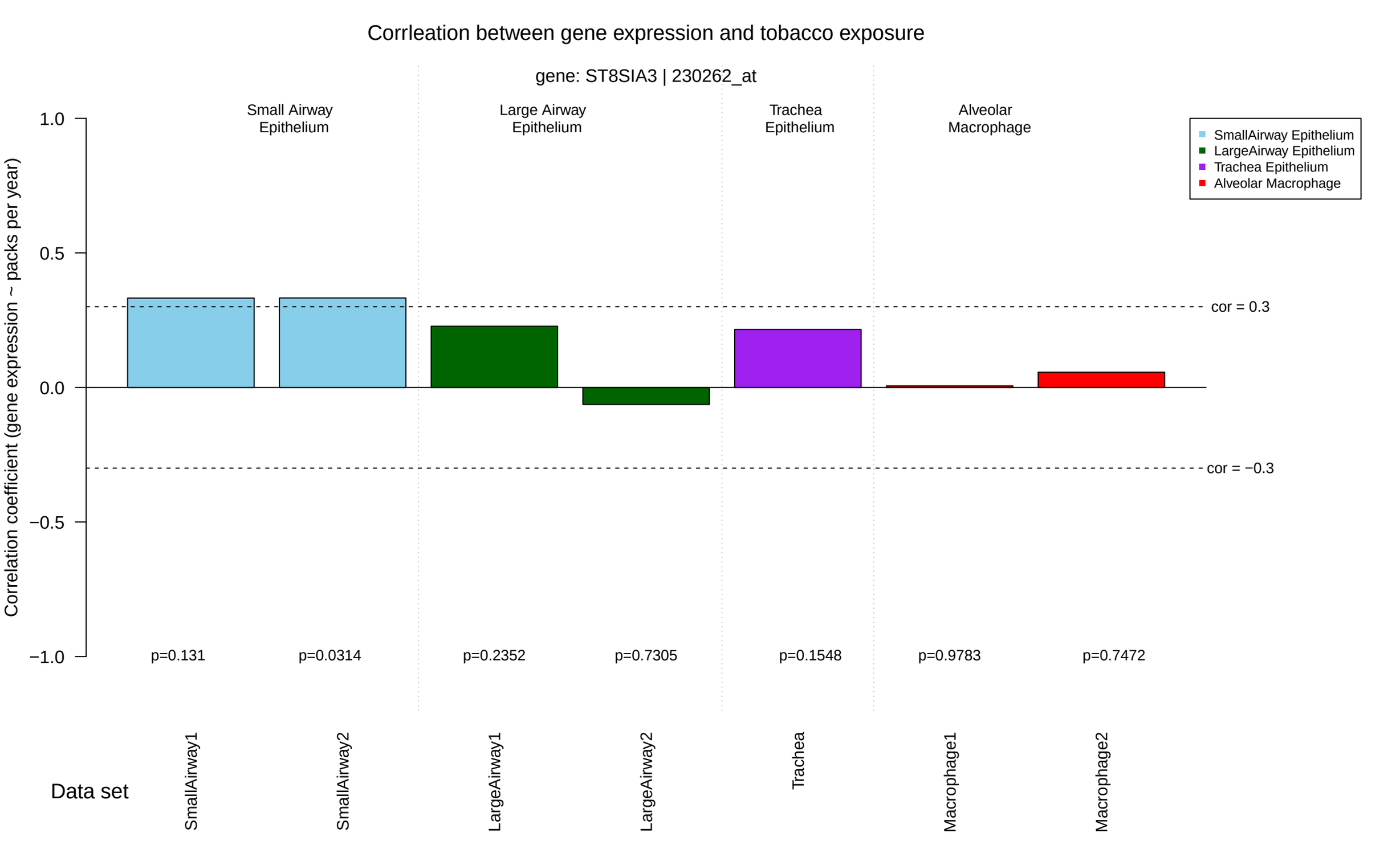

set: LargeAirway1 (p=0.0092)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3921)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0161)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9798)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8552)
ST8SIA3 | 230262_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA3 | 230262_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.131
p=0.0314
p=0.2352
p=0.7305
p=0.1548
p=0.9783
p=0.7472
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2